Algorithm Notations:

2x2x2 Specific

1. RD/ RU: Right Side Down/ Up
2. LD/ LU: Right Side Down/ Up
3. UR/ UL: Up Side Right/ Left Finger
4. DR/ DL: Down Side Right/ Left Finger
5. FR/ FL: Front Side Right/ Left Finger
6. BR/ BL: Back Side Right/ Left Finger
7. 2 after a letter: Move that 2 times (in any direction)

3x3x3 Additional:

1. MU/ MD: Middle Layer Up/ Down (Vertical Middle Line)
2. MR/ ML: Middle Layer Right/Left (Horizontal Middle Line) Finger
3. w after a letter (Wide): Move 2 Layers from that side together

Moving Entire Cube:

1. xU/ xD: Rotating the cube Up/ Down
2. yR/ yL: Rotating the cube sideways Right/ Left
3. zC/ zA: Rotating the cube Clockwise/ Anticlockwise

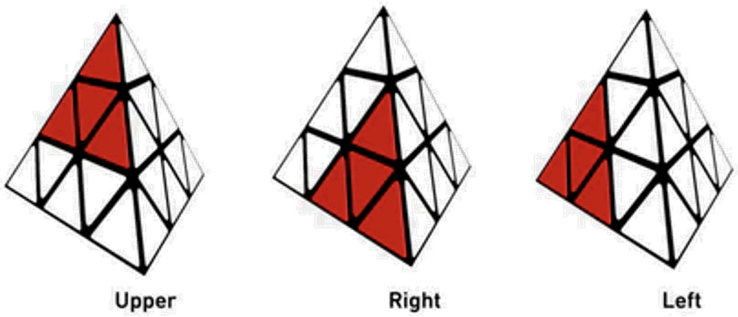
4x4x4 Additional:

1. 2 before a letter: Only move the second layer of that side (inner side)

5x5x5 - 7x7x7 Additional:

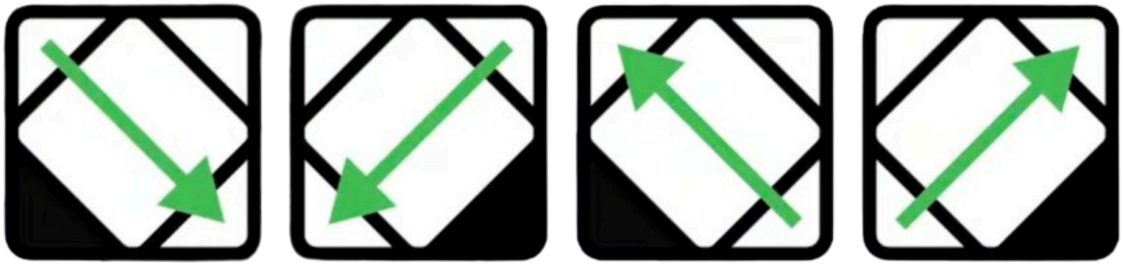
1. A number (2/3/4…) before a letter and w after a letter: Move that many layers from that side (including the edges)

Pyraminx:



Master Pyraminx:
A Master Pyraminx has the same moves as a Megaminx, just it can have w (Wide) moves too. For example, RU and RwU.

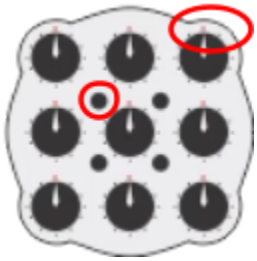
Skewb:



These moves mean the bottom corners (ones in black) are kept stationary, while moving the top half. There can also be moves where the top corners are kept stationary, while moving the bottom halves.

Clock Cube:
It has 2 types of parts that can be moved. The 4 circular objects (one is circled) in the middle are called pins that can be pushed up or down. On the 4 corners there are dials that can be rotated.

- If a pin is pushed up, and its corresponding dial is rotated, all the clocks around that pin will move (others will not).
- If 2 pins are pushed and any one of the corresponding dials is rotated, all the clocks around both the pins will move. Same for 3 and 4 pins pushed up.



2x2x2 Cube

Average Time to Solve: 30 seconds

Step 1: Solve the white color. Keep the White color facing up, and use the following algorithm:

RD, DR, RU, LD, DL, LU, RD, DR, RU

Step 2: Keep the White facing down. Now we solve the yellow color, on the face pointing up. Place the yellow pattern in the following way.

- Yellow Dot: place it on bottom left
- Yellow Line: place it on the right
- Yellow Diagonals: place one on bottom left and the other on top right
- No Yellow on top: place however

Step 3: Use the following algorithm. Repeat Steps 2 and 3 until Yellow is solved.

RU, UR, RD, UR, RU, U2, RD

Step 4: See if there is a 3rd color getting solved, and face it away from you. White is still at bottom and yellow on top. Use the following algorithm:

RD, FR, RD, B2, RU, FL, RD, B2, R2

If there was no color solved after Step 3, use the Step 4 algorithm, and one color gets solved. Then place the color away from you and repeat Step 4.

Tips to Save Time

Average Time to Solve: 15 seconds

Tip 1: Solve the White side, with the first layer aligned, all other colors on the white pieces align properly to form the first layer.

Tip 2: To solve the yellow side, use one of the 7 OLL (Orientation of Last Layer) patterns:



1. R2, U2, RU, U2, R2
2. RU, U2, R2, UL, R2, UL, R2, U2, RU
3. RU, U2, RD, UL, RU, UL, RD
4. RU, UR, RD, UR, RU, U2, RD
5. FR, RD, FL, RU, UR, RU UL, RD
6. RU, UR, RD, UL, RD, FR, RU, FL
7. FR, RU, UR, RD, UL, FL

Tip 3: For the remaining solve of the cube, don't solve a 3rd color and turn the entire cube. Instead, if a color line is solved, just put that at the back, use the solving algorithm, and solve the cube by rotating the Up layer in the end.

Tip 4: If no color is solved for the last layer of the cube, use the following algorithm. A color will be solved on the left, so just do UR, and repeat the algorithm to solve the cube.

RD, FR, RD, B2, RU, FL, RD, B2, R2

Tip 5: Don't always solve white first. Solve whichever color has the best alignment after the scramble, then do its opposite color, then the rest. Opposite pairs:

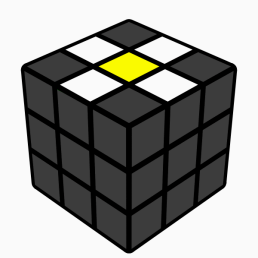
- White - Yellow
- Blue - Green
- Red - Orange

3x3x3 Cube

Average Time to solve: 1 minute

Step 1: White Cross

Solve Daisy. The Daisy is a shape with the yellow center, surrounded by white edges. The corner pieces are irrelevant in this scenario.



Keep the Yellow center facing up, rotate the Up layer until the second color of the white edge matches with its center (red, green, blue, orange) then F2 to get that edge on the white center. Do this for each of the 4 white edges.

Step 2: First Layer

Get the perfect white corner in its position (Between red and green, a corner piece with white, red and green will go). This solves the white and the entire first layer.

Step 3: Second Layer

Keep the White facing down. From the top layer, align an edge without yellow to its center. The edge will have to go to the left or the right (as shown in image).



If the edge moves to the left (as shown in image):

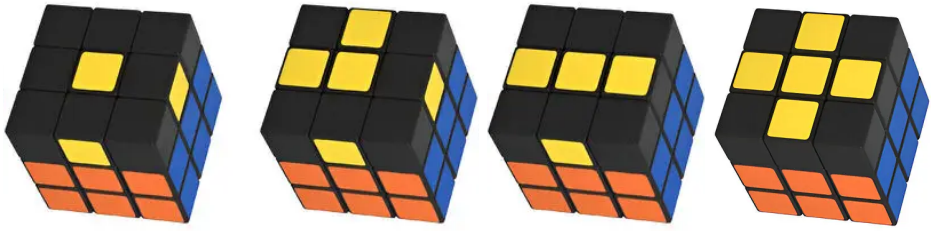
UL, LU, UR, LD, UR, FR, UL, FL

If the edge moves to the right (opposite of image):

UR, RU, UL, RD, UL, FL, UR, FR

Step 4: Yellow Cross

On the top face (Yellow centered), there should be 4 possibilities with regards to the yellow center, and yellow edges.



Use this algorithm to go from the dot (3 times), to the L (2 times), to the line (1 time), to the cross (0 times, skip this step if you get a cross).

FR, RU, UR, RD, UL, FL

Step 5: Yellow Color

Keep the White facing down. Now we solve the yellow color. Place the yellow pattern in the following way.

- Yellow Cross + 1 corner: place corner on bottom left
- Yellow Cross + Line: place line on the right
- Yellow Cross + Diagonals (8 shape): place one on bottom left and the other on top right

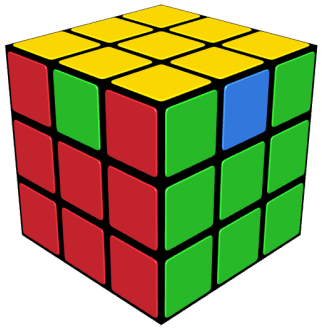
Use the following algorithm. Repeat this step until Yellow is solved.

RU, UR, RD, UR, RU, U2, RD

Step 6: Last Layer

For the last layer of the cube, you should have 6 scenarios: (With Scenarios 4, 5, and 6 skip this step)

1) All 4 sides random pattern
2) One side headlight and 3 sides random
3) One side color solved and 3 sides random
4) All 4 sides headlights
5) 3 Sides headlights and 1 side solved
6) Full Cube Solved



Headlight shown in image (2 headlights shown)

Scenario 2 and 3: Put the Headlight/Color at the back, white facing down, yellow facing up. Use the following algorithm:

RD, FR, RD, B2, RU, FL, RD, B2, R2

Scenario 1: Use the above algorithm to get 1 solved color/headlight. This takes you to Scenario 2/ 3. Repeat Scenario 2/ 3.

Step 7: You should now have either Scenario 4, 5, or 6. Scenario 6 is a completely solved cube.

Scenario 5: Place the solve color opposite to you. The middle piece in each of the 3 headlights either goes clockwise or anticlockwise to solve the cube.

- Clockwise: M2, UL, MD, U2, MU, UL, M2
- Anticlockwise: M2, UR, MD, U2, MU, UR, M2

Scenario 4: Use any clockwise/ anticlockwise algorithm on any side to reach Scenario 5.

Tips to Save Time

Average Time to Solve: 40 seconds

Tip 1: Don't solve the white cross with a daisy. Align the other color of the white edge to its centre and solve the white cross (aligned) directly.

Tip 2: After putting one white corner in its correct place, put the edge that is supposed to go there before moving to another corner. The edges are easier to move around if there are no other white corners placed restricting movement.

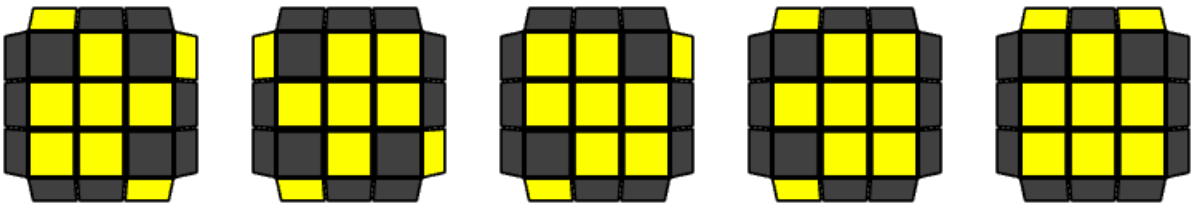
Try basic F2L (First 2 Layer) Techniques to put both the white corner and the edge in its place together to save time wherever opportunity is given.

Tip 3: To get from the Yellow L to the Yellow cross by skipping the Yellow line, use the following algorithm:

FwR, RU, UR, RD, UL, FwL

To go from the dot, use the algorithm to go to L then use the above algorithm.

Tip 4: To solve the Yellow side, use the basic OLL (Orientation of Last Layer) patterns:



1. RU, UR, RD, UR, RU, U2, RD
2. RU, U2, RD, UL, RU, UL, RD
3. FR, RD, FL, RwU, UR, RU, UL, RwD
4. RwU, UR, RD, UL, RwD, FR, RU, FL
5. R2, DR, RU, U2, RD, DL, RU, U2, RU

Tip 5: If you get 4 headlights, where each edge piece has to go to the opposite side (Orange edge in red headlights and vice versa, blue edge in green headlights and vice versa) use the algorithm:

M2, UL, M2, U2, M2, UL, M2

Tip 6: If a headlights/ 3rd color is solved, don't solve that side and shift the whole cube, instead, just rotate the Up layer to make the headlight/ color face back, solve the scenario, then rotate the Up layer to solve the cube.

Tip 5: Don't always solve white first. Solve whichever color has the best alignment after the scramble, then do its opposite color, then the rest. Opposite pairs:

- White - Yellow
- Blue - Green
- Red - Orange

Extra: A mirror cube is the same as a 3x3. Instead of having different colors, there are different sizes. Each piece is unique, with no 2 pieces having the same size. It is solved exactly like a 3x3, where instead of color alignment, the size is aligned.

4x4x4 Cube

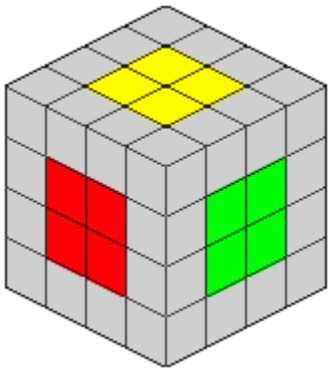
Average Time to Solve: 3 minutes

Step 1: White Center
The goal is to make a white 2x2 in the middle of any selected face. The edges/corners are irrelevant here. Solve it like you would a 2x2x2 cube or 3x3x3 cube.

Step 2: Yellow Center
On the side opposite to the white center, you need to make a yellow center without disrupting the white center. To put a yellow piece on top without disrupting a white center:

- If there is a yellow piece on top, position the yellow piece in the middle so it is corresponding to the one on top. (Top is yellow side)
- Move the Right/Left layer to bring it up into the yellow piece on top. Move the Up layer so that a random center piece is in the position that the previous yellow was.
- Move the Right/Left layer back down to bring the yellow down.

Step 3: All Other Centers
Make all the other centers in the middle. Use the technique to move a piece into a corresponding piece, moving the up layer to put a random piece in the way then moving the previous piece back.



Position the centers exactly how a 3x3 would be. If the white center is facing up:

- Orange is to the right of Blue
- Green is opposite Blue
- Red is to the left of Blue

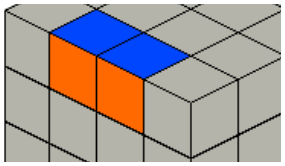
To check where each center is positioned, corner pieces are always correctly aligned. If you put a white, blue and orange corner with the white on the white center side, orange is to the right of blue.

At the end, the 4x4 cube should look like this (the other pieces are irrelevant).

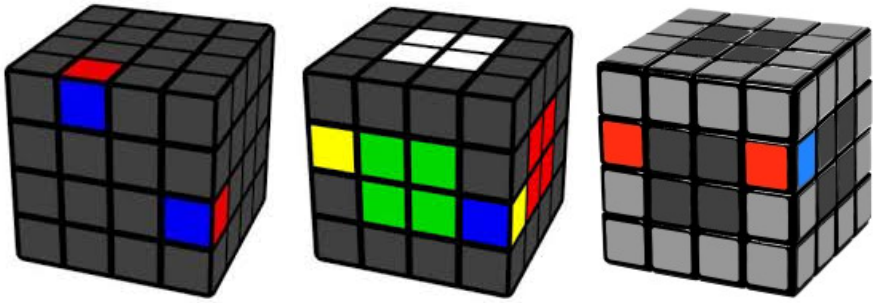
If you position centers wrong and need to swap them, use the following algorithms:

Adjacent Centers: RwU, U2, RwD, LwU, U2, LwD
Opposite Centers: Rw2, U2, Rw2, Lw2, U2, Lw2

Step 4: Edge Pairing
The goal is to pair all the edges of the same colors together, as shown in the image:



Find 2 edges that can be paired together, and get them on the same face. There are 3 scenarios possible:



Scenario 1: Edges are on adjacent sides.
Scenario 2: Edges on opposite sides, with opposite colors on the same face.
Scenario 3: Edges on opposite sides, with the same color on the same face.

For Scenario 1, keep one edge on top, the other on the right (like in image) and do UL, LD. This takes you to either Scenario 2 or Scenario 3.

For Scenario 2, move Uw (left or right) to align both the edges. To avoid this spoiling a center, move the edge into the upper layer by using RU/LU, move U to put a random unfixed edge into the previous position of the edge pair, bring it down with RD/LD, and fix the centers again.

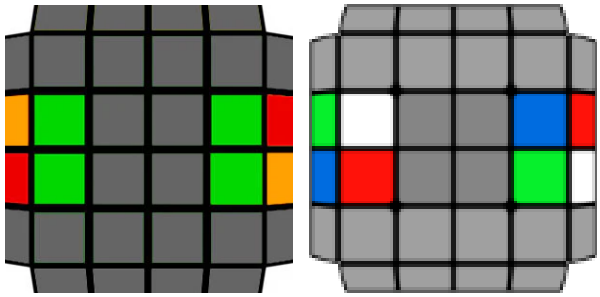
For Scenario 3, flip the edge to get Scenario 2 using the flipping algorithm:

Flipping Algorithm (Right Side): RU, UR, RD, FR, RD, FL, RU
Flipping Algorithm (Left Side): LU, UL, LD, FL, LD, FR, LU

1. Do the white edge pairs first, put them on the white side, then keep the white face down.
2. Do the remaining edge pairs without moving the white side edge pairs out of their place.
3. Put finished edge pairs on the yellow side, and make new edge pairs in the middle.

Step 5: Last 2 Edge Pairs

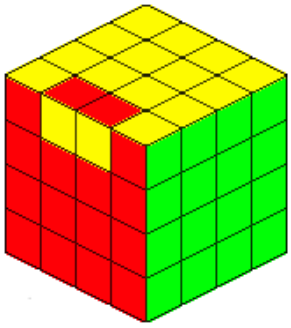
When there are 2 edge pairs remaining, place them on the same face opposite to each other. There are 2 scenarios in which they can be:



For Scenario 2, use the flipping algorithm to get to Scenario 1. Then, use the algorithm:
UwL, Right Flipping Algorithm, UwR

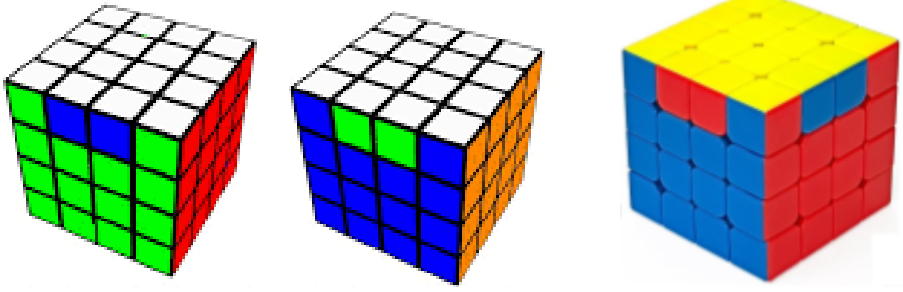
Step 6: Solve it like a 3x3x3 cube.
There are some cases that a 3x3x3 cube does not have, but a 4x4x4 can have when solving it like a 3x3x3. They are called Parities.

OLL Parity: One yellow edge is flipped. Keep the flipped side facing you.



RwU, U2, xU, RwU, U2, RwU, U2, RwD, U2, LwD, U2, RwD, U2, RwU, U2, RwD, U2, RwD
(xU means rotate up, so the side that was down first is facing you now)

PLL Parity: Headlights positioned wrong. There are 2 scenarios.



Scenario 1 (First 2 images are 1 Scenario): 2R2, U2, 2R2, Uw2, 2R2, Uw2
Scenario 2 (Third Image): Use the same algorithm as scenario 1, to get a regular headlights case.

5x5x5 - 7x7x7 Cube

Average Time to Solve:

- 5x5x5 Cube: 6 minutes
- 6x6x6 Cube: 10 minutes
- 7x7x7 Cube: 15 minutes

Step 1: White and Yellow Centers

Same procedure as a 4x4x4 for these 2 centers

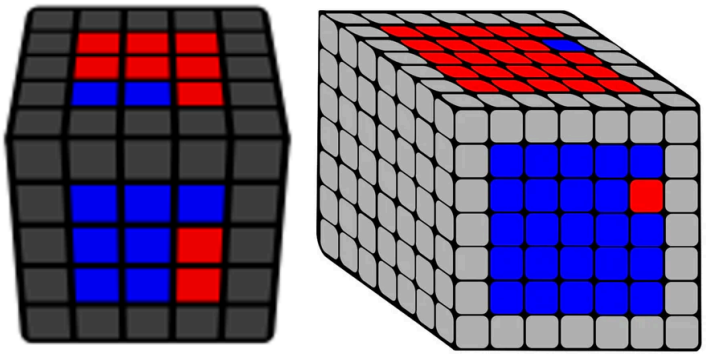
Step 2: Adjacent Center Pair

Similar to 4x4x4. In a 4x4x4, you can choose any order for the remaining centers. However, in 5x5x5, 6x6x6, and 7x7x7 you should do adjacent pairs of centers first. Then, the last 2 centers will be an adjacent pair too. Adjacent pairs are:

- Blue - Orange
- Blue - Red
- Green - Orange
- Green - Red

Step 3: Last 2 Centers

For the last 2 centers, there can be singular or more center pieces that are not present in the correct place. They also cannot be put in the adjacent center without spoiling other fully made centers. (Examples shown in images)



To solve this:

- Put the unsolved center piece in the front face
- Put the location where it needs to go in the top face.
- Align them horizontally (the second image shows vertical alignment which is wrong).
- Move the right/left inner slice up to insert the piece.
- Turn the U layer so the inserted piece is no longer in the slice that will be moved back down.

- Move the new slice up again.
- Undo the U-turn in Step 4.
- Bring the first slice back down.
- Repeat the Step 4 U-turn.
- Bring the second slice back down.
- Repeat until the last two centers are solved.

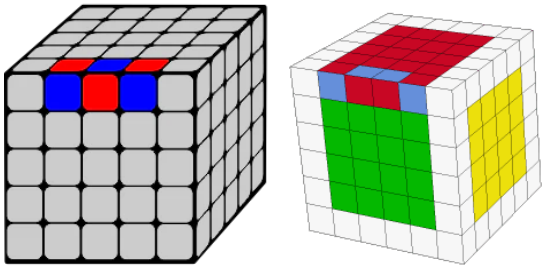
5x5x5 and 7x7x7 have fixed centers, where it is not possible to swap centers. 6x6x6 can have swapped centers. To swap centers in case of mistakes:

Adjacent Centers: 3RwU, U2, 3RwD, 3LwU, U2, 3LwD
Opposite Centers: 3Rw2, U2, 3Rw2, 3Lw2, U2, 3Lw2

Step 4: Edge Pairing

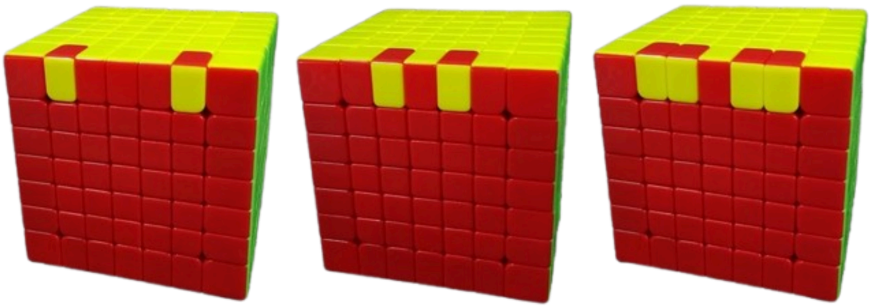
This is done exactly like how it is done in a 4x4x4 cube. This includes the last 2 edges. Where you position them to have the same color on one face, then put it in place using the flipping algorithm.

However, there are cases for the 5x5, 6x6, 7x7 where the last edge can have flipped pieces. This is called edge parity.



5x5 Edge Parity: RwU, U2, xU, RwU, U2, RwU, U2, 3RwD, U2, LwD, U2, RwD, U2, RwU, U2, RwD, U2, RwD
6x6 Edge Parity: RwU, U2, xU, RwU, U2, RwU, U2, RwD, U2, LwD, U2, 4RwD, U2, RwU, U2, RwD, U2, RwD

For a 7x7, there are 3 different edge parity scenarios:



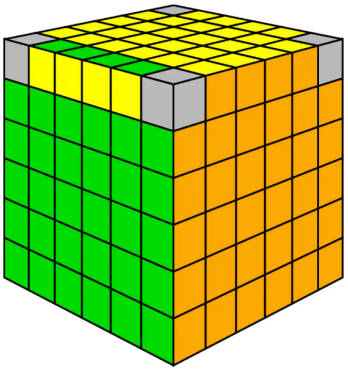
Scenario 1: RwU, U2, xU, RwU, U2, RwU, U2, RwD, U2, LwD, U2, 5RwD, U2, RwU, U2, RwD, U2, RwD

Scenario 2: Use same as Scenario 1. This will take you to Scenario 3.

Scenario 3: 3 RwU, U2, xU, 3RwU, U2, 3RwU, U2, 3RwD, U2, 3LwD, U2, 4RwD, U2, 3RwU, U2, 3RwD, U2, 3RwD

Step 5: Solve like a 3x3x3 Cube

A 6x6x6 is an even cube, so like the 4x4x4 it can have an OLL and a PLL parity, while the 5x5x5 and 7x7x7 can only have an edge parity.



6x6x6 OLL Parity: 3RwU, U2, xU, 3RwU, U2, 3RwU, U2, 3RwD, U2, 3LwD, U2, 3RwD, U2, 3RwU, U2, 3RwD, U2, 3RwD

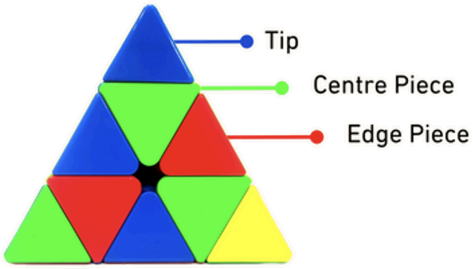


6x6x6 PLL Parity: 3Rw2, FR2, U2, 3Rw2, R2, U2, F2, 3Rw2

Pyraminx

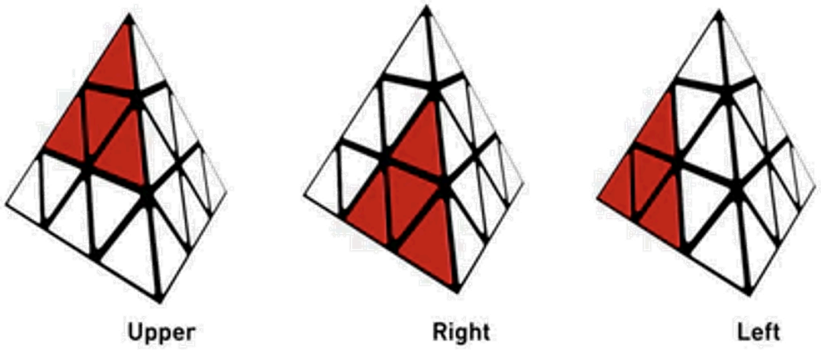
Average Time to Solve: 25 seconds

Types of Pieces:
There are 3 types of pieces:



The centers are arranged in a way that there is only one face where all 3 centers of the same color come together. That face is also the only one where all 3 tips of the same color come together.

Types of Moves:



Step 1: Aligning the Centers and Tips
Choose a color to start with. Find the face where all 3 of the centers and tips of that color could be on. Rotate the centers and tips to align on that face.

Step 2: Completing the First Layer
The edges of the first layer can be in 3 positions: (Hold with the color chosen on the bottom)

- In the second layer of the cube.
- In the first layer of the cube, but in the wrong place (flipped or normal).
- In the first layer of the cube and in the correct place.

For the first scenario, rotate the Up layer so that the edge to be inserted is on the same face (directly above) the position it needs to be inserted in. Keep in mind that the color being solved (on the bottom) is the one facing you.

Depending on which side the edge is, use the following algorithm:

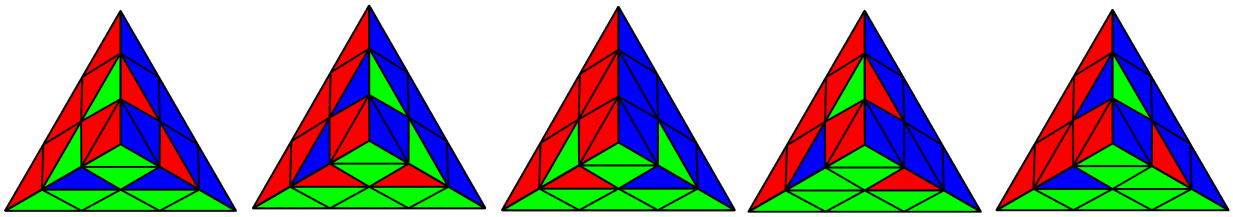
Left Side: LD, RD, LU, RU

Right Side: RD, LD, RU, LU

For the second scenario, use any of the above 2 algorithms once to get the edge in the second layer, then repeat the first scenario steps. Repeat with all 3 edges.

Step 3: Last Layer Cases

Below is the top view, all of them are solved with a front view hold (facing green face in image)



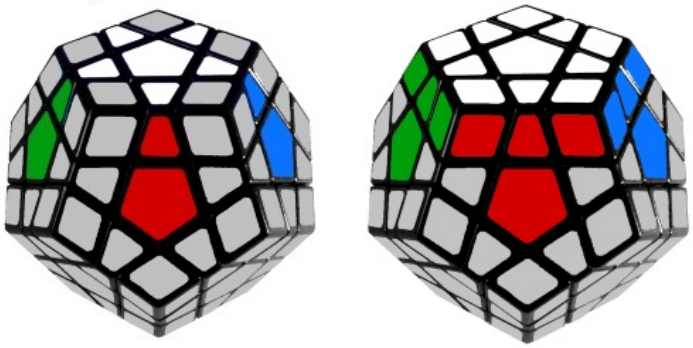
1. Headlights (shifting right): RU, UR, RD, UR, RU, UR, RD
2. Headlights (shifting left): RU, UL, RD, UL, RU, UL, RD
3. Two edges flipped: LD, RD, LU, RU, UL, RU, UR, RD
4. Each edge shifting right: RD, LD, RU, LU, UL, LU, UR, LD
5. Each edge shifting left: LD, RD, LU, RU, UR, RU, UL, RD

Megaminx

Average Time to Solve: 4 minutes

Step 1: White Star + First Layer

Each face on the megaminx has 5 sides, so instead of a cross, we make a star, with each color aligned to the center (shown in first image). Then, put the correct corner in its place, forming the first layer. This step is similar to the 3x3x3.



Step 2: First 2 Layers

Like the 3x3x3, you have to put the colored edges (like red-blue or red-green) in its correct position. Use the same algorithms as the 3x3x3 to put the edges. Here, the edge to be inserted must be on the side adjacent to the location it needs to be inserted. If it is 2 sides away, it will not work.

Step 3: Solving the First Layer of colors

On a regular megaminx, the first layer of colors is blue, red, green, purple, and yellow. The goal is to solve all of them completely, and with perfect alignment to the second layer of colors' centers.



1. Choose any of the first layer colors to start with. Let's assume we start with yellow.
2. By rotating unsolved layers, make the yellow star, with each edge aligned to the color.
3. Insert the yellow-blue corners and yellow-purple corner, but using F2L techniques (First 2 layers), to insert a blue and purple edge with the corner itself.

4. For the yellow corner in the middle, you can insert it normally. This completes the yellow color completely.
5. Move clockwise, (purple next) or anticlockwise (blue next) and repeat the process. But do all the colors in circular order, without skipping any.

You can use a 4x4x4 flipping algorithm to flip edge orientations. To insert F2L without spoiling any solved colors, insert the corner + edge from the last layer of the megaminx, the one with the grey center. Your final result should look like the image above.

Step 4: Second 2 Layers

Now, you must insert the edges of the last layer of colors, which are light blue, cream, pink, orange, and light green. The process for this is the same as the 3x3x3, and the same as inserting the edges in step 2.



The cream-light blue edge will be inserted between the 2 colors, in the empty spot below.

Step 5: The Gray Star

There are 4 different scenarios:



The 4th is a complete star, in which case skip the step. Position the cases as shown in the image. The gray is facing up and you are looking at the cube from the bottom side of the image.

Case 1: FR, RU, UR, RD, UL, FL
Case 2: FR, UR, RU, UL, RD, FL
Case 3: Use any of the algorithms above to take you to one of the above cases.

Step 6: Aligning the Gray Star

Rotate the gray layer to see the maximum number of edges that are aligned. If there are 2 edges aligned, and both edges are beside each other, hold the cube so that gray is on top, one aligned edge is facing you, and the other aligned edge is on your left.

Then, use the algorithm: (repeat one more time if needed)

RU, UR, RD, UR, RU, UL2, RD

If 2 edges aligned are opposite, or 2 sides apart, turn the layer clockwise (UR), and see if one edge gets solved. If not, turn the layer anticlockwise (UL). One edge has to be solved in either of the 2 rotations. Face that solved edge in front of you and do the algorithm once or twice (same as above):

RU, UR, RD, UR, RU, UL2, RD

This should take you to the first case. If only one edge is aligned no matter the rotation, then do the algorithm above once from any alignment which will take you to either of the 2 above cases.

Step 7: The Gray Face

Turn the cube so that the gray side is at the bottom. Pick one gray corner that is not solved, and use the algorithm:

RU, UR, RD, UL

Use this 2 or 4 times to flip the gray corner towards the gray face. Even if the gray corner gets flipped and inserted after RD, remember to complete the algorithm with a UL.

Then, rotate the bottom layer so that another unflipped gray corner is in the place of the first one. Do not rotate the cube, rotate only the bottom layer. Repeat the algorithm and rotate until every gray corner is flipped and the gray face is solved.

The rest of the solved portion of the cube should return to its original state, which is an indication you did it correctly.

Step 8: Aligning Gray Corners and Solving Megaminx

Now all that is left is putting the correct corners in the correct position. Keep gray faced down.

RU, UR, RD and RU, UL, RD

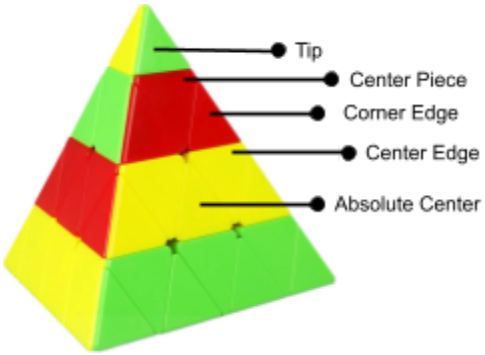
Find a corner that is not in its correct position. Remove it using the first algorithm. Rotate the bottom layer to find the location where that removed corner must be inserted. Similar to the previous step, do not rotate the entire cube or any other layer. Insert the corner with the 2nd algorithm.

By inserting the corner, you removed an incorrectly aligned corner simultaneously. Rotate to find its position, use the first algorithm to insert it. Repeat this process until all the corners are correctly positioned. The megaminx should be completely solved.

Master Pyraminx

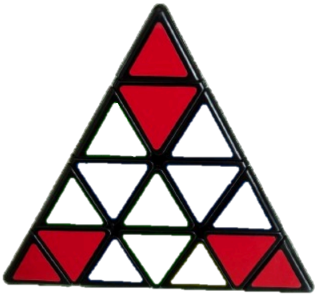
Average Time to Solve: 3 minutes

Types of Pieces:

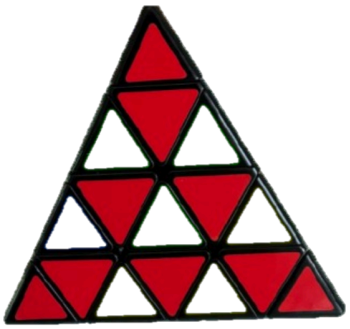


Types of Moves:
A Master Pyraminx has the same moves as a Megaminx, just it can have w (Wide) moves too. For example, RU and RwU.

Step 1: Aligning Tips and Centers
This is exactly like a Pyraminx. Choose a color to start with, for example, red. Find the face where all 3 of the centers and tips of that color could be on. Rotate the centers and tips to align on that face.



Step 2: Aligning Center edges
This is similar to inserting edges in a Pyraminx. Face the red on the bottom. Rotate the up layer so the Center Edge to be inserted is above the location where it must be inserted. The red side of the center edge must be facing you in this case. Use the algorithm:



If Center Edge is on Left: LwD, RwD, LwU, RwU
If Center Edge is on Right: RwD, LwD, RwU, LwU

The final result should look like this image. The Absolute Center must be ignored for now.

Step 3: First Layer
Now you have to insert all the Corner edge pieces. Face the red down. Rotate a Corner Edge until it is above the location it must be inserted into. If the edge is in the 2nd layer, use the following algorithm:

If Corner Edge is on Left: UwL, LU, UwR, LD
If Corner Edge is on Right: UwR, RU, UwL, RD

If the edge is in the 3rd layer, replace Uw with U, and L/R with Lw/Rw. Use the same logic of the above algorithm. The first full layer must be solved. The Absolute Center can be red or any other color, it will be solved in the end.

Step 4: Second Layer Center Edges
Face the red at the bottom. Now, you have to align the 2 center edges for each of the 3 other faces (blue, green, and yellow). The cases for this are exactly like the Last Layer Cases in the Pyraminx. The only difference is you must use w (Wide) moves instead of normal.

Refer to Page 15 for this. Then, twist the Up layer until the Center Piece and tip for each of the 3 colors is aligned.

Step 5: 2nd Layer Corner Edges
The goal is to insert edges from the third layer into the second layer to solve them. Even though a master pyraminx has 2 Corner Edges of the same color, one Corner Edge cannot go in the place of the other.

- Find an unsolved corner edge location in the second layer, and the respective corner edge in the third layer. For example, a random edge is in the place of blue - green, and the blue - green corner edge is in the third layer.
- Rotate the tip and the third layer, so that the Corner edge from the third layer is on the same face as its location. If you have the blue side facing you (indicated by the first layer), then make sure the blue side of the Edge is facing you.
- We need the Edge (3rd layer) and its location to be on opposite sides. For example, if the edge is on the right side, the location must be on the left, and vice versa. However, the color of the edge cannot be switched, the blue side of the edge should still be on the blue face.
- If they both are on the same side, that is the wrong edge, so find another edge and location.
- If the Edge (3rd Layer) is on the left, and the location is on the right, use the algorithm:
  RD, LwD, RU, LwU

Notice how this spoils the Corner edge in the red layer. This can be inserted very easily by:

UL, LwU, UR, LwD

- If the Edge (3rd Layer) is on the right, and the location is on the left, use the algorithm (reverse):

LD, RwD, LU, RwU

To Re-insert Removed Red Edge: UR, RwU, UL, Rwd

- Repeat this with every edge in the second layer.

If all the unsolved Corner edges are in the 2nd Layer, just in the wrong spots. Use one of the above algorithms to kick out one Corner edge into the 3rd layer and repeat the step.

Step 6: Last Layer

The last layer can have 3 cases:

- Completely Solved (skip this step then)
- Headlights where each Edge needs to rotate clockwise
- Headlights where each Edge needs to rotate anticlockwise

Use the following algorithms:

Clockwise Rotation: RwD, UL, RwU, UL, RwD, UL, RwU

Anticlockwise Rotation: RwD, UR, RwU, UR, RwD, UR, RwU

Step 7: Swapping Absolute Centers and Solving Master Pyraminx

There are 4 cases you can have:

- 3 unsolved absolute centers and 1 solved center (Clockwise rotation)
- 3 unsolved absolute centers and 1 solved center (Anticlockwise rotation)
- 4 unsolved absolute centers (2 swapped pairs)

Scenario 1: RwD, UwR, RwU, UwR, RwD, UwR, RwU
\+ RwD, UL, RwU, UL, RwD, UL, RwU (to fix last layer)
Scenario 2: RwD, UwL, RwU, UwL, RwD, UwL, RwU
\+ RwD, UR, RwU, UR, RwD, UR, RwU (to fix last layer)

It is the same algorithm as the last layer, but with Uw, instead of U, and UR and UL are swapped.

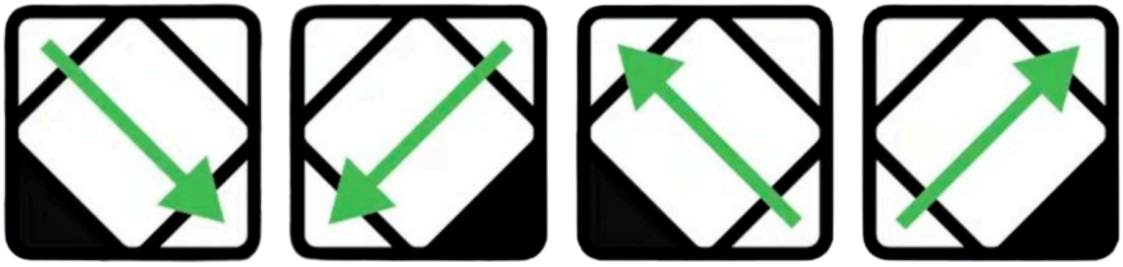
Doing this for Scenario 1 and 2 take you to Scenario 3. Hold the Master Pyraminx so that from one of the swapped absolute center pairs, one is facing down and the other is facing you. Do this algorithm:

(RwD, LwD, RwU, LwU) Repeat 3 times

Skewb

Average Time to Solve: 1 minute

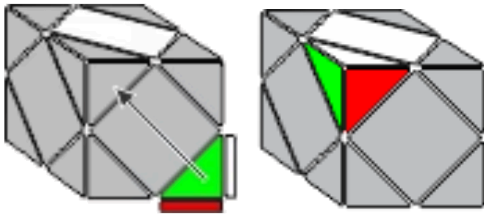
Types of Moves:



These moves mean the bottom corners (ones in black) are kept stationary, while moving the top half. There can also be moves where the top corners are kept stationary, while moving the bottom halves.

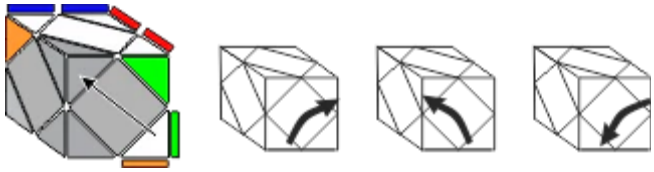
Step 1: White Face
The goal is to solve the white face with first layer corners aligned. These corners must be aligned to each other, not to their centers. The centers will be aligned later. Keep White on top

1. Find the White - Red - Green Corner, and hold it so that the corner is on the bottom right side. Twist the Right layer (RU/RD) until the white corner looks like the image below, then use LU, to insert the corner in place.

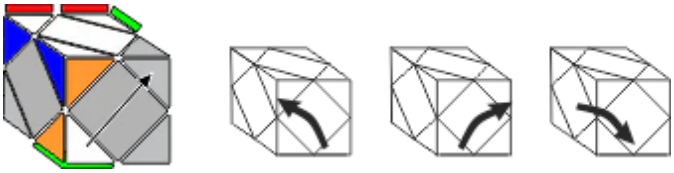


2. Turn the Skewb clockwise, and repeat the process with the White - Blue - Red corner, and then the White - Orange - Blue Corner.
3. Get the last corner, White - Green - Orange, to the bottom right, exactly like the previous 3 corners. You can have 3 scenarios: (you cannot just turn the Right side RU/RD since that removes an inserted White corner)

   Scenario 1: The corner has the White facing like the image. Just use LU to insert it.
   Scenario 2: The corner has White facing in front, like the image shown below. In that case use this algorithm: RU, LU, RD



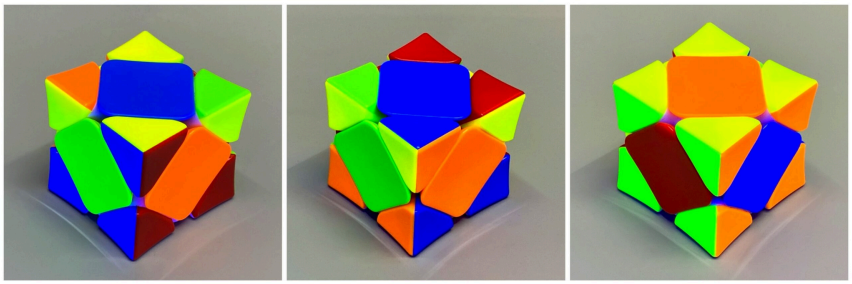
Scenario 3: The corner has White facing at the bottom. In that case, use L2, to get to the image below, then use the algorithm: LU, RU, LD



Step 2: The Yellow Corners

Hold the White facing bottom. The goal is to get 4 yellow corners facing up. There are 3 cases:

- 2 Yellow Corners facing up, 2 facing outward.
- No Yellow Corners facing up, 2 yellow corners facing outward on one face.
- 4 Yellow Corners facing up (Skip this step)



Scenario 1: Use the below algorithm from any of the 4 sides. This takes you to Scenario 2.
Scenario 2: Use the below algorithm with the 2 yellow corners on the same face facing the right side.



RD, LD, RU, LU

(The RD/RU/LD/LU is not the same as the one from Step 1, see the image to understand)

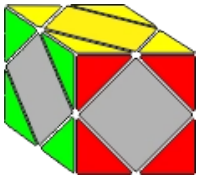
Step 3: Aligning the Yellow Center

If the Yellow Center is aligned and solved, skip this step. Otherwise, the Yellow Center will be on one of the 4 faces adjacent to white. Hold the White facing at the bottom, and the yellow center facing away from you (at the back) and repeat the same algorithm as above.



RD, LD, RU, LU

Then turn the cube (clockwise or anticlockwise) 2 times, with White still on the bottom, and repeat the algorithm above. The Yellow should be solved, and look like the image below:



Step 4: Aligning Remaining Centers and Solving Skewb

There can be 2 cases for the last 4 sides:

- One solved side, and 3 unsolved sides (just centers misaligned)
- 4 unsolved sides (2 pairs of swapped centers)

Scenario 1: Among the 3 unsolved sides, one of the sides will be such that the center color is an opposite of the corner colors. Opposite pairs are the same as 3x3:

- Red - orange
- Blue - green

Hold the Skewb so that the White face and the Yellow face are on the right and left sides (can be either), and the face with the opposite pair is facing you. Repeat the same algorithm as Step 3:



RD, LD, RU, LU

Then turn the cube (clockwise or anticlockwise) 2 times, and repeat the algorithm above. This solves the Skewb.

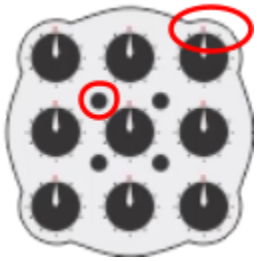
Scenario 2: Hold the Skewb with any of the 4 colors facing you, White and Yellow facing on the left and right sides, and use the same algorithm as Scenario 1, turn the cube clockwise/ anticlockwise twice and repeat the algorithm.

This takes you to Scenario 1, where you keep the opposite pair facing you and use the algorithm.

Clock Cube

Average Time to Solve: 1 minute

How is it solved:
A clock cube is solved when each hand on each clock on both sides (usually white and black) points straight up. This is shown in the image.

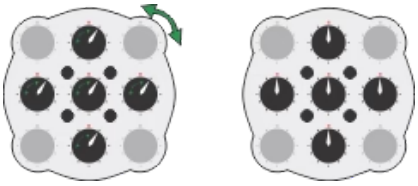


How a Clock Cube works:
It has 2 types of parts that can be moved. The 4 circular objects (one is circled) in the middle are called pins that can be pushed up or down. On the 4 corners there are dials that can be rotated.

- If a pin is pushed up, and its corresponding dial is rotated, all the clocks around that pin will move (others will not).
- If 2 pins are pushed and any one of the corresponding dials is rotated, all the clocks around both the pins will move. Same for 3 and 4 pins pushed up.

Step 1: Solving a cross
The goal is to have all the clocks forming a + aligned. For that, you have to match the center to 1 edge clock, then both those to another edge clock, then all 3 to a third edge clock, then to the last.

- Keep the bottom right pin pushed up, rest pushed down, and rotate until the center clock matches the top clock.
- Keep the top left pin pushed up, rest pushed down, and rotate until the top and center clock match the right clock (both the clocks will move the exact same).
- Keep the top right pin pushed up, rest pushed down, and rotate until the top, right, and center clock match the bottom clock.
- Keep the top right and bottom right pins pushed up, rest pushed down, and rotate until the other 3 clocks match the left clock.
- Keep all pins up, and turn any dial until all 5 clocks are pointing up.



Step 2: Second Cross
Repeat the Step 1 process from the other side. Here, it is very important not to rotate a dial if its corresponding pin is not pushed up. Otherwise, this will ruin the cross from the other side. For this, you do not need to get them all pointing up yet.

Step 3: Solving 4 Corners
Solving this for 1 side results in solving it for both sides. Similar to the previous steps, we are not solving the corner clocks, we are matching all the solved clocks to the corner clocks, one by one.

- Keep all the pins pushed up except the top left. Rotate a corresponding dial so that the 5 solved clocks match the top left clock.
- Keep all the pins pushed up except the top right. Rotate a corresponding dial so that the 6 solved clocks match the top right clock.
- Keep all the pins pushed up except the bottom right. Rotate a corresponding dial so that the 7 solved clocks match the bottom right clock.
- Keep all the pins pushed up except the bottom left. Rotate a corresponding dial so that the 8 solved clocks match the bottom left clock.

Step 4: Solving Clock Cube
Push all the pins up, and rotate so that each clock points up. If you did everything right, the clock cube should be completely solved, and look like the image below.

